Week 5 · Lab 2

Version C

Graded Practice:

`?:`, `switch` and `while`

10 problems from LeetCode · 10 points each · checked by the TA at your desk

Week: 5 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lecture 4 — the ternary operator `?:`, `switch`/`case`, the `while` loop, `break`/`continue`

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. There are **10 problems, 10 points each** (100 in total): problems 1–3 are **easy**, 4–7 **medium** and 8–10 **hard**. Solve them in any order; a problem with no attempt earns 0 points. There is enough work for the whole session — keep going until the end.
4. Under each title, **Practises:** says which part of Lecture 4 the problem trains.
5. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard (a list of numbers or a word is read one item at a time, because arrays and strings come later). On LeetCode itself the answer is written as a *function* inside `class Solution`; functions come next week.

Quick reference

Ternary operator `condition ? a : b` is a *value*. Put it in parentheses inside `cout`.

```
int mx = a > b ? a : b;
cout << (n % 2 == 0 ? "even" : "odd");
d = d < 0 ? -d : d;                  // |d|
```

`switch` for `int`/`char` values compared with constants. `break` leaves; without it execution falls through into the next `case`. Several labels may share one body.

```
switch (c) {
    case 'a': case 'e': case 'i':
    case 'o': case 'u': vowels++; break;
    case ' ':           spaces++; break;
    default:            others++;
}
```

`while`: initialise, test, update The test runs before every pass, so the body may run zero times.

```
int i = 1;                       // initialise
while (i <= n) {                 // test
    total += i;
    i++;                         // update
}
```

Reading *n* items without an array read one, use it, forget it:

```
int n;  cin >> n;
int best = -1000000000;
int i = 0;
while (i < n) {
    int x;  cin >> x;            // also: char c; cin >> c;
    if (x > best) { best = x; }  // running maximum
    i++;
}
```

A word given as `5 hello` is read the same way: `n`, then `n` single `char`s.

Digit by digit

```
while (n > 0) {
    int d = n % 10;              // last digit
    n /= 10;                     // drop it
}
```

`break` and `continue` `break` leaves the innermost loop (or `switch`) at once; `continue` jumps back to the test. In a `while`, do the update *before* a `continue`, or the loop never ends. A `while` inside a `while`: reset the inner loop's variables inside the outer loop.

Classic bugs a missing `break`; no update, or `while (...);` — an endless loop; `i != 10` with `i += 3`; `<` versus `<=`; a product that starts at 0; `int` overflow (use `long long`).

EASY

Problems 1–3 · 30 points

Short programs: one idea each. Test every branch, not just the examples.

1 Score of a String 10 pts

LeetCode 3110 · `leetcode.com/problems/score-of-a-string`

Practises: a `while` loop over characters, and `?:` for the absolute value.

Read n ($2 \le n \le 100$), then a word of n lowercase letters. Its *score* is the sum of $|c_i - c_{i-1}|$ over neighbouring letters, using the character codes. Print the score.

Input	Output	Why
`5 hello`	`13`	$\|101-104\| + \|108-101\| + \|108-108\| + \|111-108\| = 3+7+0+3$.
`3 zaz`	`50`	$25 + 25$.

TA: ____ / 10 pts initials ____

2 Count the Digits That Divide a Number 10 pts

LeetCode 2520 · `leetcode.com/problems/count-the-digits-that-divide-a-number`

Practises: a `while` loop over the digits of a copy.

Read `num` ($1 \le$ `num` $\le 10^9$; it has no digit 0). Print how many of its digits divide `num` (count a repeated digit every time it appears).

Input	Output	Why
`7`	`1`	
`121`	`2`	1 divides 121 (twice); 2 does not.
`1248`	`4`	

TA: ____ / 10 pts initials ____

3 Three Consecutive Odds 10 pts

LeetCode 1550 · `leetcode.com/problems/three-consecutive-odds`

Practises: a `while` loop with `continue`, and a counter that resets.

Read n ($1 \le n \le 1000$), then n positive integers. Print `true` if three *consecutive* numbers are all odd, otherwise `false`.

Input	Output	Why
`4 2 6 4 1`	`false`	
`9 1 2 34 3 4 5 7 23 12`	`true`	`5 7 23`.

Hint: an even number resets the run — `continue` *with the next number; an odd one extends it.*

TA: ____ / 10 pts initials ____

MEDIUM

Problems 4–7 · 40 points

A few cases to get right. Test the boundaries.

4 Determine if String Halves Are Alike 10 pts

LeetCode 1704 · `leetcode.com/problems/determine-if-string-halves-are-alike`

Practises: `switch` with grouped `case` labels.

Read an even n ($2 \le n \le 1000$), then a word of n letters (upper or lower case). Print `true` if its first half and its second half contain the same number of vowels (`a e i o u`, in either case), otherwise `false`.

Input	Output	Why
`4 book`	`true`	bo and ok: one vowel each.
`8 textbook`	`false`	`text` has 1, book has 2.

Hint: one `switch` *with ten* `case` *labels that share one body. The position* `i` *tells you which half you are in.*

TA: ____ / 10 pts initials ____

5 Climbing Stairs 10 pts

LeetCode 70 · `leetcode.com/problems/climbing-stairs`

Practises: a `while` loop that keeps the last two values.

A staircase has n steps ($1 \le n \le 45$). Each move climbs 1 or 2 steps. Print in how many different ways you can reach the top.

Input	Output	Why
2	2	1+1 or 2.
3	3	1+1+1, 1+2, 2+1.

Hint: the last move onto step k came from step $k-1$ or from step $k-2$. So ways(k) = ways($k-1$) + ways($k-2$). Keep only the last two values.

TA: ____ / 10 pts initials ____

6 Monotonic Array 10 pts

LeetCode 896 · `leetcode.com/problems/monotonic-array`

Practises: a loop with two `bool` flags.

Read n ($1 \le n \le 10^5$), then n integers. Print `true` if they never go down (each $\ge$ the previous one) or never go up (each $\le$ the previous one), otherwise `false`.

Input	Output	Why
`4 1 2 2 3`	`true`	
`4 6 5 4 4`	`true`	
`3 1 3 2`	`false`	Up, then down.

Hint: keep two `bool`*s, "could still be non-decreasing" and "could still be non-increasing". Each pair of neighbours may switch one of them off.*

TA: ____ / 10 pts initials ____

7 Binary Gap 10 pts

LeetCode 868 · `leetcode.com/problems/binary-gap`

Practises: a `while` loop over the binary digits (`% 2`, `/ 2`).

Read n ($1 \le n \le 10^9$). In its binary form, look at each pair of *neighbouring* 1s and the distance between their positions. Print the largest such distance, or 0 if n has fewer than two 1s.

Input	Output	Why
`22`	`2`	$22 = 10110_2$: distances 2 and 1.
`8`	`0`	1000_2: only one 1.
`5`	`2`	101_2.

Hint: `n % 2` *is the last binary digit and* `n / 2` *drops it, exactly like* `% 10` *and* `/ 10` *for decimal digits. Count the positions, and remember where the previous 1 was.*

TA: ____ / 10 pts initials _____

HARD

Problems 8–10 · 30 points

More cases and more care. Read the hint before you start.

8 Sum of Number and Its Reverse 10 pts

LeetCode 2443 · leetcode.com/problems/sum-of-number-and-its-reverse

Practises: a `while` loop inside a `while` loop, and `break`.

Read `num` ($0 \le$ `num` $\le 10^5$). Print `true` if `num` $= k +$ reverse(k) for some $k \ge 0$, where reverse drops leading zeros (reverse(140) = 41); otherwise `false`.

Input	Output	Why
`443`	`true`	172 + 271.
`63`	`false`	
`181`	`true`	140 + 41.

Hint: try every k from 0 to `num`*; the inner loop reverses k.*

TA: ____ / 10 pts initials ____

9 Convert a Number to Hexadecimal 10 pts

LeetCode 405 · leetcode.com/problems/convert-a-number-to-hexadecimal

Practises: two `while` loops, and `?:` to choose a digit character.

Read an `int` `num`. Print it in hexadecimal with lowercase letters and no leading zeros. A negative number is written in 32-bit two's complement, i.e. as `num` $+2^{32}$.

Input	Output	Why
`26`	`1a`	$1 \cdot 16 + 10$.
`-1`	`ffffffff`	$-1 + 2^{32} = 2^{32} - 1$.

Hint: work in a `long long`*. Without arrays you cannot store the digits and print them backwards, so find the largest power of 16 that is $\le$ the number first, then print digits from that power down. A digit $d \ge 10$ is the letter* `'a' + d - 10`*.*

TA: ____ / 10 pts initials ____

10 Distribute Candies to People 10 pts

LeetCode 1103 · leetcode.com/problems/distribute-candies-to-people

Practises: a `while` loop inside a `while` loop.

n people stand in a row. You give 1 candy to the first, 2 to the second, . . . , n to the last, then $n + 1$ to the first, $n + 2$ to the second, and so on, until the candies run out; the last person served gets whatever is left. Read `candies` ($1 \le$ `candies` $\le 10^9$) and n ($1 \le n \le 1000$) and print how many candies each person gets, in order, separated by spaces.

Input	Output	Why
`7 4`	`1 2 3 1`	Gifts 1, 2, 3, then only 1 is left.
`10 3`	`5 2 3`	Gifts 1, 2, 3, then 4 to the first.

Hint: without arrays you cannot keep n totals at once. Instead, for each person in turn, replay the whole distribution and add up only what that person receives. Gift number t (counting from 0) goes to person $t \bmod n$.

TA: ____ / 10 pts initials ____

Score sheet — Version C

	Problem	Level	Points	Earned	TA
1	Score of a String (LC 3110)	Easy	10		
2	Count the Digits That Divide a Number (LC 2520)	Easy	10		
3	Three Consecutive Odds (LC 1550)	Easy	10		
4	Determine if String Halves Are Alike (LC 1704)	Medium	10		
5	Climbing Stairs (LC 70)	Medium	10		
6	Monotonic Array (LC 896)	Medium	10		
7	Binary Gap (LC 868)	Medium	10		
8	Sum of Number and Its Reverse (LC 2443)	Hard	10		
9	Convert a Number to Hexadecimal (LC 405)	Hard	10		
10	Distribute Candies to People (LC 1103)	Hard	10		
	Total		**100**		

After the lab

Every problem here is on LeetCode under the number shown. Here the input is read one item at a time; on LeetCode it arrives as an array or a string. Once functions are covered, rewrite your solutions as a LeetCode `class Solution` and submit them.

End of Week 5 · Lab 2 · Version C